



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

● Experiment No.: 2

1. Title: Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

2. Software/Tools Required: Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: `var` is function-scoped, `let` is block-scoped, and `const` is used for fixed values.

The `prompt()` function accepts user input, and `Number()` converts input into numeric values.

The program calculates the total amount, GST, and final bill using arithmetic operations.

An object stores all billing details together, while object destructuring extracts required values easily.

Template literals (``) are used to display formatted output with variable values using `$${}`.

The `document.write()` method displays the billing receipt on the webpage.

4. Output :

The image shows two screenshots of a web browser displaying a 'Billing Calculator' application. The browser tabs include 'sec_appti_24-28: Attendance' and 'Billing Calculator'. The address bar shows the file path: '/Users/pratham/jsexp2/shreya1196.html'.

The first screenshot shows a prompt box titled 'This page says' with the text 'Enter Customer Name :'. The input field contains 'Shreya Maske'. There are 'Cancel' and 'OK' buttons.

The second screenshot shows a similar prompt box titled 'This page says' with the text 'Enter Product Name :'. The input field contains 'Microwave'. There are 'Cancel' and 'OK' buttons.



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The screenshot shows a web browser with two tabs: 'sec_apti_24-28: Attendance' and 'Billing Calculator'. The address bar shows the file path '/Users/pratham/jsexp2/shreya1196.html'. A dialog box titled 'This page says' is open, asking 'Enter Quantity :'. The input field contains the value '1'. There are 'Cancel' and 'OK' buttons at the bottom of the dialog.

The screenshot shows the same web browser with the 'Billing Calculator' tab active. A dialog box titled 'This page says' is open, asking 'Enter Price per Item :'. The input field contains the value '42000'. There are 'Cancel' and 'OK' buttons at the bottom of the dialog.

The screenshot shows the 'Shreya's Billing Calculator' web application. The title is 'Shreya's Billing Calculator'. Below it is the heading 'Billing Reciept' (note the typo). The receipt details are as follows:

- Customer: Shreya Maske
- Product: Microwave
- Quantity: 1
- Price per item: 42000
- Total: ₹42000
- GST (18%): ₹7560.00
- Final Bill: ₹49560.00**

5. **Result/Conclusion:** A billing calculator was successfully created using JavaScript. The program accepted user input, calculated the bill with GST, and demonstrated the use of var, let, const, template literals, and object destructuring effectively.



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Case Study No.: 2

1. **Case Study Title:** Develop the webpage for an online shopping website with using html,css and html tags and calculate bill,customer name,mobile no,invoice no,product no,quantity(kg),rate per kg,discount,GST,packing charges,payment mode(cash/card/UPI), Membership(yes/no).

2. Case Study Program Code:

```
shreya1196.html case2.html X
case2.html > html > body > script > bill
1 <!DOCTYPE html>
2 <html>
3 <head>
4 | <title>Online Shopping Bill</title>
5 </head>
6 <body style="background-color: ■ rgb(230, 218, 173);">
7 <h1>Shreya's Online Shopping Website</h1>
8 <script>
9 const GST = 0.18;
10 // Customer Details
11 var custname = prompt("Enter Customer Name:");
12 var mobile = prompt("Enter Mobile Number:");
13 var invoice = prompt("Enter Invoice Number:");
14 var product = prompt("Enter Product Name:");
15 var quantity = Number(prompt("Enter Quantity (Kg):"));
16 var rate = Number(prompt("Enter Rate per Kg:"));
17 var discount = Number(prompt("Enter Discount (%):"));
18 var packing = Number(prompt("Enter Packing Charges:"));
19 var payment = prompt("Enter Payment Mode (Cash/Card/UPI):");
20 var membership = prompt("Membership (Yes/No):");
21 // Calculations
22 let total = quantity * rate;
23 let discountAmt = total * discount / 100;
24 let subtotal = total - discountAmt;
25 let gstAmt = subtotal * GST;
26 let finalBill = subtotal + gstAmt + packing;
```



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```
27 // Object
28 const bill = {
29     customer: custname,
30     mobile,
31     invoice,
32     product,
33     quantity,
34     rate,
35     discount,
36     packing,
37     payment,
38     membership,
39     total,
40     discountAmt,
41     gstAmt,
42     finalBill
43 };
44 // Destructuring
45 const {
46     customer,
47     mobile: mob,
48     invoice: inv,
49     product: pname,
50     payment: payMode,
51     membership: member,
52     gstAmt: gst,
53     finalBill: amount
54 } = bill;
55 // Output
56 document.write(`<h2>Shopping Invoice</h2>`);
57 document.write(`<hr>`);
58 document.write(`<p><b>Customer Name:</b> ${customer}</p>`);
59 document.write(`<p><b>Mobile No:</b> ${mob}</p>`);
60 document.write(`<p><b>Invoice No:</b> ${inv}</p>`);
61 document.write(`<p><b>Product:</b> ${pname}</p>`);
62 document.write(`<p><b>Quantity:</b> ${quantity} Kg</p>`);
63 document.write(`<p><b>Rate per Kg:</b> ₹${rate}</p>`);
64 document.write(`<p><b>Total:</b> ₹${total.toFixed(2)}</p>`);
65 document.write(`<p><b>Discount:</b> ₹${discountAmt.toFixed(2)}</p>`);
66 document.write(`<p><b>GST (18%):</b> ₹${gst.toFixed(2)}</p>`);
67 document.write(`<p><b>Packing Charges:</b> ₹${packing.toFixed(2)}</p>`);
68 document.write(`<p><b>Payment Mode:</b> ${payMode}</p>`);
69 document.write(`<p><b>Membership:</b> ${member}</p>`);
70 document.write(`<h2>Final Bill: ₹${amount.toFixed(2)}</h2>`);
71 </script>
72 </body>
73 </html>
```



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3. Output:

← → ↻ ⓘ File /Users/shreyamaske/jsexp2/case2.html

☐☐

Shreya's Online Shopping Website

Shopping Invoice

Customer Name: Shreya Maske

Mobile No: 7798642334

Invoice No: 880024

Product: Basmati Rice

Quantity: 6 Kg

Rate per Kg: ₹120

Total: ₹720.00

Discount: ₹14.40

GST (18%): ₹127.01

Packing Charges: ₹35.00

Payment Mode: Card

Membership: No

Final Bill: ₹867.61

4. Result/Conclusion: The online shopping billing webpage was successfully developed using **HTML, CSS, and JavaScript**. The program accepted customer and product details, calculated the total bill including **discount, GST, and packing charges**, and displayed the final invoice correctly. It demonstrated the use of user input, arithmetic calculations, objects, template literals, and object destructuring in JavaScript.